MATH 411: Honors Complex Variables

Midterm 2 Review Problems – Solutions

Fall 2026 | Midterm 2: Friday, November 6, 2026, 3:00–3:50 PM, CAB 369

These problems are for practice. They cover Lectures 13–23 (Lang §§3.3–3.7, 4.1), as described on the Midterm 2 review sheet. The exam is closed book, so practise writing complete solutions without notes, stating each result you use and checking its hypotheses.

Try each problem yourself before reading its solution.

1. (Cauchy estimates and growth.)

(a) State the Cauchy estimates for $f^{(n)}(z_0)$, including the hypotheses.

Solution

If f is holomorphic on an open set U with $\overline{D(z_0, R)} \subseteq U$ and $\|f\|_R = \max_{|z-z_0|=R} |f(z)|$, then $|f^{(n)}(z_0)| \le n!\,\|f\|_R / R^n$ for all $n \ge 0$.

(b) Let f be entire with $|f(z)| \le 2 + 3|z|^{5/2}$ for all $z \in \mathbb{C}$. Prove that f is a polynomial of degree at most 2.

Solution

For every $R > 0$ and $n \ge 3$, the Cauchy estimates on $|z| = R$ give

$$|f^{(n)}(0)| \le \frac{n!\,(2 + 3R^{5/2})}{R^n} = n!\Big(\frac{2}{R^n} + \frac{3}{R^{n-5/2}}\Big) \xrightarrow[R\to\infty]{} 0,$$

since $n - \frac{5}{2} > 0$. So $f^{(n)}(0) = 0$ for $n \ge 3$. As f is entire, its Taylor series at 0 converges to f on all of $\mathbb{C}$, so $f(z) = f(0) + f'(0)z + \frac{1}{2}f''(0)z^2$.

(c) Is there an entire function with $|f(z)| \le 1 + |z|^{1/2}$ for all z and $f(i) \ne f(0)$?

Solution

No. The same argument with $n \ge 1$ (now $n - \frac{1}{2} > 0$) gives $f^{(n)}(0) = 0$ for all $n \ge 1$, so f is constant and $f(i) = f(0)$.

2. (Liouville's theorem.)

(a) Let f be entire with $\operatorname{Re} f(z) \le 4$ for all z. Prove that f is constant.

Solution

Let $g = e^f$, which is entire with $|g(z)| = e^{\operatorname{Re} f(z)} \le e^4$. By Liouville, g is constant, so $0 = g' = f' e^f$. Since e^f never vanishes, $f' \equiv 0$, and since $\mathbb{C}$ is connected, f is constant. (Alternatively apply Liouville to $1/(5 - f)$, which is bounded by 1.)

(b) Let f be entire with $f(z+1) = f(z)$ and $f(z+i) = f(z)$ for all z. Prove that f is constant.

Solution

f is continuous on the compact square $S = \{x + iy : 0 \le x, y \le 1\}$, so $|f| \le M$ on S. Given $z = x + iy$, put $w = z - \lfloor x \rfloor - i\lfloor y \rfloor \in S$. Repeated use of the periodicity gives $f(z) = f(w)$, so $|f| \le M$ on $\mathbb{C}$. By Liouville, f is constant.

(c) Use Liouville's theorem to prove that every nonconstant polynomial has a root in $\mathbb{C}$.

Solution

Let $P(z) = a_n z^n + \cdots + a_0$ with $n \ge 1$, $a_n \ne 0$, and suppose P has no root. Then $1/P$ is entire. Since $|P(z)| \ge |a_n||z|^n\big(1 - \sum_{k<n} \frac{|a_k|}{|a_n||z|^{n-k}}\big) \ge \frac{|a_n|}{2}|z|^n$ for $|z| \ge R_0$, $1/P$ is bounded on $|z| \ge R_0$, and it is bounded on the compact disc $|z| \le R_0$. By Liouville, $1/P$ is constant, so P is constant, a contradiction.

3. (Laurent series on different annuli.) Let $f(z) = \dfrac{1}{z(z-2)}$.

(a) Find the Laurent expansion of f on $0 < |z| < 2$.

Solution

Partial fractions give $f(z) = \frac{1}{2}\big(\frac{1}{z-2} - \frac{1}{z}\big)$. For $|z| < 2$, $\frac{1}{z-2} = -\sum_{n\ge 0} \frac{z^n}{2^{n+1}}$, so on $0 < |z| < 2$

$$f(z) = -\frac{1}{2z} - \sum_{n\ge 0} \frac{z^n}{2^{n+2}}.$$

(b) Find the Laurent expansion of f on $|z| > 2$.

Solution

For $|z| > 2$, $\frac{1}{z-2} = \frac{1}{z} \cdot \frac{1}{1-2/z} = \sum_{n\ge 1} 2^{n-1} z^{-n}$. Hence

$$f(z) = \frac{1}{2}\Big(\sum_{n\ge 1} 2^{n-1} z^{-n} - z^{-1}\Big) = \sum_{n\ge 2} 2^{n-2} z^{-n}.$$

(c) Compute $\int_{|z|=1} f(z)\,dz$ and $\int_{|z|=3} f(z)\,dz$ (counterclockwise) using (a) and (b).

Solution

By the coefficient formula $a_{-1} = \frac{1}{2\pi i}\int_{|z|=\rho} f$, valid for any circle in the annulus: $\int_{|z|=1} f = 2\pi i \cdot (-\frac{1}{2}) = -\pi i$, and $\int_{|z|=3} f = 2\pi i \cdot 0 = 0$. The two expansions differ because they are Laurent series on *different* annuli. Uniqueness holds only on a fixed annulus.

4. (More Laurent series.) Compute the indicated Laurent expansion and use it to evaluate the integral (all circles counterclockwise).

(a) $z^2 e^{1/z}$ on $0 < |z| < \infty$; $\displaystyle\int_{|z|=1} z^2 e^{1/z}\,dz$.

Solution

$z^2 e^{1/z} = \sum_{n\ge 0} \frac{z^{2-n}}{n!} = z^2 + z + \frac{1}{2} + \frac{1}{6z} + \frac{1}{24z^2} + \cdots$. The z^{-1} coefficient is $\frac{1}{3!} = \frac{1}{6}$, so the integral is $2\pi i/6 = \pi i/3$.

(b) $\dfrac{1}{z^2(1-z)}$ on $0 < |z| < 1$; $\displaystyle\int_{|z|=1/2} \frac{dz}{z^2(1-z)}$.

Solution

$\frac{1}{z^2(1-z)} = \sum_{n\ge 0} z^{n-2} = z^{-2} + z^{-1} + 1 + z + \cdots$, so $a_{-1} = 1$ and the integral is $2\pi i$.

(c) $\dfrac{1}{z^2(1-z)}$ on $|z| > 1$; $\displaystyle\int_{|z|=2} \frac{dz}{z^2(1-z)}$.

Solution

For $|z| > 1$, $\frac{1}{1-z} = -\frac{1}{z}\cdot\frac{1}{1-1/z} = -\sum_{n\ge 1} z^{-n}$, so $\frac{1}{z^2(1-z)} = -\sum_{n\ge 3} z^{-n}$. Here $a_{-1} = 0$, so the integral is 0.

5. (Logarithms and roots.)

(a) Let U be simply connected and f holomorphic and nowhere zero on U. Prove that there is a holomorphic g on U with $g^3 = f$.

Solution

By the theorem on logarithms of nonvanishing functions, there is a holomorphic L on U with $e^L = f$ (take $L(z) = w_0 + \int_{z_0}^{z} f'/f$ with $e^{w_0} = f(z_0)$, which is well defined because U is simply connected). Then $g = e^{L/3}$ satisfies $g^3 = e^L = f$.

(b) Prove that there is no continuous $g : \mathbb{C}\setminus\{0\} \to \mathbb{C}$ with $g(z)^3 = z$.

Solution

Let $h(t) = g(e^{it})e^{-it/3}$ for $t \in [0, 2\pi]$. Then h is continuous and $h^3 \equiv 1$, so h takes values in the finite set $\{1, \omega, \omega^2\}$ with $\omega = e^{2\pi i/3}$. By connectedness of $[0, 2\pi]$, h is constant. But $h(2\pi) = g(1)e^{-2\pi i/3} = \omega^{-1}h(0) \neq h(0)$, a contradiction.

(c) With Log the principal branch, compute $\mathrm{Log}\big((-1+i)^2\big)$ and $2\,\mathrm{Log}(-1+i)$. Explain the discrepancy.

Solution

$(-1+i)^2 = -2i$, so $\mathrm{Log}(-2i) = \ln 2 - \frac{\pi i}{2}$. Also $-1+i = \sqrt{2}\,e^{3\pi i/4}$, so $2\,\mathrm{Log}(-1+i) = \ln 2 + \frac{3\pi i}{2}$. These differ by $2\pi i$. The value $2\,\mathrm{Arg}(-1+i) = 3\pi/2$ lies outside $(-\pi, \pi]$, so $z \mapsto 2\,\mathrm{Log}\, z$ is a logarithm of z^2, but not the principal one. In general $L_{f_1 f_2} = L_{f_1} + L_{f_2}$ holds only up to $2\pi i\mathbb{Z}$.

6. (Shifting a rectangular contour.) Let $a > 0$ and $b \in \mathbb{R}$. Using $\int_{-\infty}^{\infty} e^{-ax^2}\,dx = \sqrt{\pi/a}$, prove that

$$\int_{-\infty}^{\infty} e^{-ax^2}\cos(bx)\,dx = \sqrt{\frac{\pi}{a}}\,e^{-b^2/(4a)}.$$

Hint: integrate e^{-az^2} around the rectangle with vertices $\pm R$, $\pm R + ic$ for a suitable c.

Solution

Since the integrand is even in b and the case $b = 0$ is given, assume $b > 0$, and let $c = \frac{b}{2a}$. Let Q_R have vertices $-R, R, R + ic, -R + ic$. Since e^{-az^2} is entire, $\int_{\partial Q_R} e^{-az^2} dz = 0$.
Vertical sides: for $z = \pm R + iy$ with $0 \le y \le c$, $|e^{-az^2}| = e^{-a(R^2 - y^2)} \le e^{ac^2 - aR^2}$. Each side has length c, so by ML each contributes at most $c\, e^{ac^2 - aR^2} \to 0$.
Top: $-a(x + ic)^2 = -ax^2 + ac^2 - 2iacx = -ax^2 + \frac{b^2}{4a} - ibx$. Traversed from right to left, the top contributes $-e^{b^2/(4a)} \int_{-R}^{R} e^{-ax^2} e^{-ibx} dx$.
Letting $R \to \infty$ gives $\int_{-\infty}^{\infty} e^{-ax^2} e^{-ibx} dx = e^{-b^2/(4a)} \sqrt{\pi/a}$. The sine part vanishes because its integrand is odd, so taking real parts gives the claim.

7. (A sector contour: Fresnel integrals.) Integrate e^{iz^2} over the boundary of the sector $\{re^{i\theta} : 0 \le r \le R,\ 0 \le \theta \le \pi/4\}$ to prove that

$$\int_0^{\infty} \cos(t^2)\, dt = \int_0^{\infty} \sin(t^2)\, dt = \frac{\sqrt{2\pi}}{4}.$$

You may use $\int_0^{\infty} e^{-r^2} dr = \frac{\sqrt{\pi}}{2}$ and the inequality $\sin 2\theta \ge \frac{4\theta}{\pi}$ for $0 \le \theta \le \frac{\pi}{4}$.

Solution

Since e^{iz^2} is entire, Cauchy's theorem gives $0 = I_1 + I_2 + I_3$, where:
$I_1 = \int_0^R e^{it^2} dt$ (the segment $[0, R]$).
I_2 is the arc $z = Re^{i\theta}$, $0 \le \theta \le \pi/4$. Here $|e^{iz^2}| = e^{-R^2 \sin 2\theta} \le e^{-4R^2\theta/\pi}$, so

$$|I_2| \le \int_0^{\pi/4} e^{-4R^2\theta/\pi} R\, d\theta \le R \cdot \frac{\pi}{4R^2} = \frac{\pi}{4R} \to 0.$$

I_3 is the ray $z = re^{i\pi/4}$, traversed inward. Since $iz^2 = i \cdot r^2 i = -r^2$, we get $I_3 = -e^{i\pi/4} \int_0^R e^{-r^2} dr$. Letting $R \to \infty$ gives $\int_0^{\infty} e^{it^2} dt = e^{i\pi/4} \frac{\sqrt{\pi}}{2} = \frac{\sqrt{\pi}}{2\sqrt{2}}(1 + i)$; in particular this limit exists. Taking real and imaginary parts, both integrals equal $\frac{\sqrt{\pi}}{2\sqrt{2}} = \frac{\sqrt{2\pi}}{4}$.

8. (An indented semicircle.) Prove that $\lim_{R \to \infty} \int_0^R \frac{\sin x}{x}\, dx = \frac{\pi}{2}$ by integrating e^{iz}/z over the closed contour made of $[-R, -\varepsilon]$, the upper semicircle $|z| = \varepsilon$ (clockwise), $[\varepsilon, R]$, and the upper semicircle C_R of radius R (counterclockwise). *Hints:* write $\frac{e^{iz}}{z} = \frac{1}{z} + E(z)$ with E entire, and use $\sin\theta \ge \frac{2\theta}{\pi}$ on $[0, \frac{\pi}{2}]$.

Solution

The contour lies in $U = \mathbb{C} \setminus \{iy : y \le 0\}$, which is star-shaped about i and hence simply connected. Since e^{iz}/z is holomorphic on U, its integral over the contour is 0.
Real segments: $\int_{-R}^{-\varepsilon} \frac{e^{ix}}{x} dx + \int_{\varepsilon}^{R} \frac{e^{ix}}{x} dx = \int_{\varepsilon}^{R} \frac{e^{ix} - e^{-ix}}{x} dx = 2i \int_{\varepsilon}^{R} \frac{\sin x}{x} dx$.
Small semicircle ($z = \varepsilon e^{i\theta}$, θ from π to 0): $\int \frac{dz}{z} = \int_{\pi}^{0} i\, d\theta = -\pi i$. Since $E(z) = \frac{e^{iz} - 1}{z}$ is entire, it is bounded by some M near 0, so $|\int E| \le M\pi\varepsilon \to 0$.

Large semicircle: $|e^{iz}| = e^{-R\sin\theta}$ and $|dz/z| = d\theta$, so

$$\Big|\int_{C_R} \frac{e^{iz}}{z}dz\Big| \le \int_0^{\pi} e^{-R\sin\theta}d\theta = 2\int_0^{\pi/2} e^{-R\sin\theta}d\theta \le 2\int_0^{\pi/2} e^{-2R\theta/\pi}d\theta \le \frac{\pi}{R} \to 0.$$

Letting $\varepsilon \to 0$ and $R \to \infty$ gives $2i\int_0^{\infty}\frac{\sin x}{x}dx - \pi i = 0$, so the integral is $\frac{\pi}{2}$.

9. (Cauchy's integral formula.) Compute, with all circles counterclockwise:

(a) $\displaystyle\int_{|z|=2} \frac{e^z}{(z-1)^2}\,dz$

Solution

By Cauchy's formula for derivatives with $f = e^z$ and $z_0 = 1$: $\int \frac{f(z)}{(z-1)^2}dz = 2\pi i f'(1) = 2\pi i e$.

(b) $\displaystyle\int_{|z|=2} \frac{\sin z}{z^4}\,dz$

Solution

Here $\frac{f^{(3)}(0)}{3!} = \frac{1}{2\pi i}\int \frac{f}{z^4}$ with $f = \sin z$ and $f'''(0) = -\cos 0 = -1$, so the integral is $2\pi i \cdot (-\frac{1}{6}) = -\frac{\pi i}{3}$.

(c) $\displaystyle\int_{|z|=1} \frac{\cos z}{z(z-3)}\,dz$

Solution

$g(z) = \frac{\cos z}{z-3}$ is holomorphic on $D(0,3) \supseteq \overline{D(0,1)}$, so the integral is $2\pi i\, g(0) = -\frac{2\pi i}{3}$.

(d) $\displaystyle\int_{|z|=3} \frac{dz}{z^2+1}$

Solution

$\frac{1}{z^2+1} = \frac{1}{2i}(\frac{1}{z-i} - \frac{1}{z+i})$, and each term integrates to $2\pi i$ around $|z| = 3$. So the integral is $\frac{1}{2i}(2\pi i - 2\pi i) = 0$.

10. (Morera, primitives, and homotopy.)

(a) State Morera's theorem and the homotopy form of Cauchy's theorem.

Solution

Morera: if f is continuous on an open U and $\int_{\partial Q} f = 0$ for every closed rectangle $Q \subseteq U$, then f is holomorphic on U. *Homotopy form:* if f is holomorphic on U and γ, η are curves in U that are homotopic in U with fixed endpoints (or closed curves that are homotopic through closed curves), then $\int_\gamma f = \int_\eta f$. In particular, $\int_\gamma f = 0$ for every null-homotopic closed curve γ in U.

(b) Let f_n be holomorphic on an open set U and suppose $f_n \to f$ uniformly on U. Prove that f is

holomorphic on U.

Solution

f is continuous, being a uniform limit of continuous functions. For a closed rectangle $Q \subseteq U$, Cauchy–Goursat gives $\int_{\partial Q} f_n = 0$, so by ML, $|\int_{\partial Q} f| = |\int_{\partial Q}(f - f_n)| \le L(\partial Q) \sup_U |f - f_n| \to 0$. Hence $\int_{\partial Q} f = 0$, and Morera's theorem shows f is holomorphic.

(c) Show that $\int_\gamma z^{-2}\,dz = 0$ for every closed piecewise C^1 curve γ in $\mathbb{C} \setminus \{0\}$, but that the unit circle is *not* null-homotopic in $\mathbb{C} \setminus \{0\}$.

Solution

z^{-2} has the primitive $-z^{-1}$ on $\mathbb{C} \setminus \{0\}$, so $\int_\gamma z^{-2} dz = -\gamma(b)^{-1} + \gamma(a)^{-1} = 0$ for closed γ. On the other hand, $1/z$ is holomorphic on $\mathbb{C} \setminus \{0\}$ and $\int_{|z|=1} dz/z = 2\pi i \neq 0$. If the circle were null-homotopic there, the homotopy form of Cauchy's theorem would force this integral to be 0.